Munim Thahmid

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Experience

Research Intern May 2026 - Present
University of Illinois Urbana-Champaign (UIUC) *Remote*

- Building **TLAPS-Bench** [link], a 956-task benchmark across 71 TLA+ system specifications for evaluating frontier LLMs on mechanically verified proof completion and generation.
- Evaluated GPT-5.6 Sol, GPT-5.5, Claude Opus 4.8, and Gemini 3.1 Pro across agentic and single-turn settings; measured GPT-5.6 Sol at 99.0% vs. 18.8% task pass on a matched 293-task slice and analyzed sensitivity to reasoning effort, specification complexity, proof scaffolding, and iterative verifier feedback.
- Designed strict proof-completion and proof-from-scratch evaluators with read-only contexts, marked edit regions, canonical replay, anti-cheating gates, and TLAPM verification; built reproducible runners with reasoning controls, retries, resumability, and token and cost telemetry across six agent backends.
- Building **SREGym** [link], a 90-problem live benchmark for LLM agents; extended its evaluation stack with independent agent and judge endpoints, state-based recovery oracles, and a 9-question **LLM-as-a-Judge** rubric validated against experts ($\kappa = 0.90$) for controlled clean and noisy incidents.

Software Developer, AI Systems Sep 2024 - Feb 2026
Yobo AI *Remote*

- Built FastAPI and asynchronous backend services, agent workflows, streaming telephony integrations, and automated testing infrastructure for production AI voice-agent applications.

Projects

Script Matters / BanglishBench [link] | *B.Sc. Thesis* 2026

- Built a controlled multilingual reliability benchmark with aligned Bangla, Banglish, and English views; preservation constraints, human review, and automated validation produced 1,174 usable paired items.
- Benchmarked 3 local Qwen models and 5 hosted frontier LLMs, including GPT-5.5 and Sonnet 4.6, under frozen prompts and parsers; showed that high overall accuracy can conceal script-conditioned fragility and traced failures through tokenization, recoverability, and format compliance.
- Fine-tuned Qwen2.5-3B with Hugging Face PEFT/LoRA on Banglish-only and balanced multilingual data using source-disjoint splits; evaluated robustness mitigation with paired bootstrap intervals and McNemar tests.

Publication

Whisper Encoder Representations for Bengali Phone-like Units | *First Author* INTERSPEECH 2026

- Built a PyTorch, Hugging Face, and TorchAudio pipeline for layer-wise representations from Whisper-small, medium, large-v3, and self-supervised XLS-R; trained supervised linear and MLP probes on speaker-disjoint data.
- Characterized model-scale and representation behavior with three-seed experiments, macro-F1, ABX, label-shuffle and text-disjoint controls, ablations, and speaker-level bootstrap confidence intervals.

Education

Bangladesh University of Engineering and Technology (BUET) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering | CGPA: 3.46/4.00 *Jan 2022 - May 2026*

Technical Skills

Formal Methods and Evaluation: TLA+, TLC, TLAPS/TLAPM, formal specifications, model checking, theorem proving, LLM benchmarks
ML and LLMs: PyTorch, Hugging Face Transformers, PEFT/LoRA, TorchAudio, Qwen, Whisper, XLS-R, LiteLLM
Programming and Systems: Python, C/C++, Linux, Git, Docker, Kubernetes, FastAPI, Bash, PostgreSQL